



Artificial Intelligence Foundations (CIS4049-N)

Week 3 – Machine Learning – An Introduction – Part 1

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Today's plan

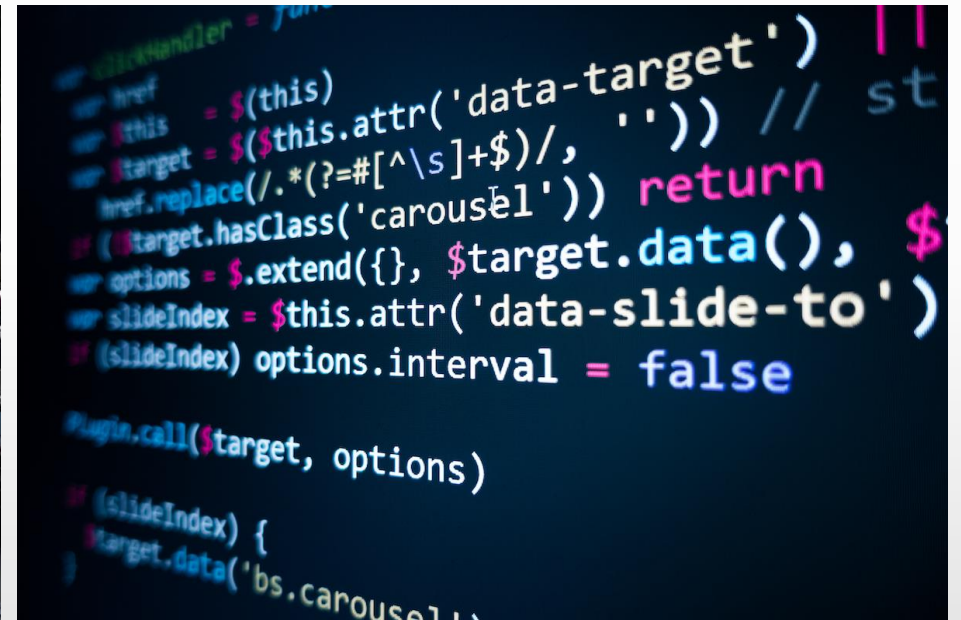
- **Pillars of AI**
- **What is Machine Learning?**
- **Machine Learning - Categories**



Credit: <https://www.intelligentcio.com/africa/2020/07/27/machine-learning-technology-garners-enterprise-thrust/>

Pillars of Artificial Intelligence

How do we solve AI tasks?



AI Paradigm – Three Pillars of AI

Modelling

Inference

Learning

AI Paradigm – Modelling

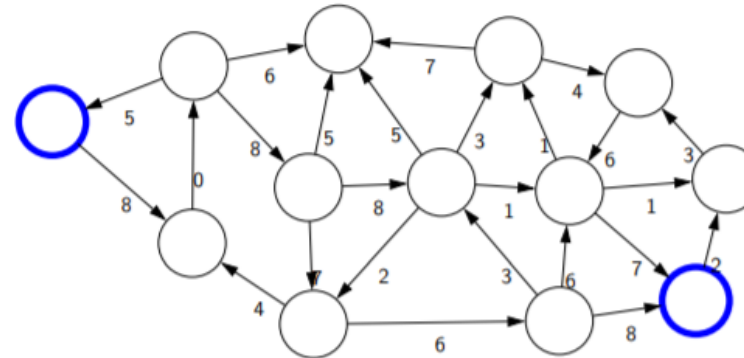
Real world



MODELLING



Model



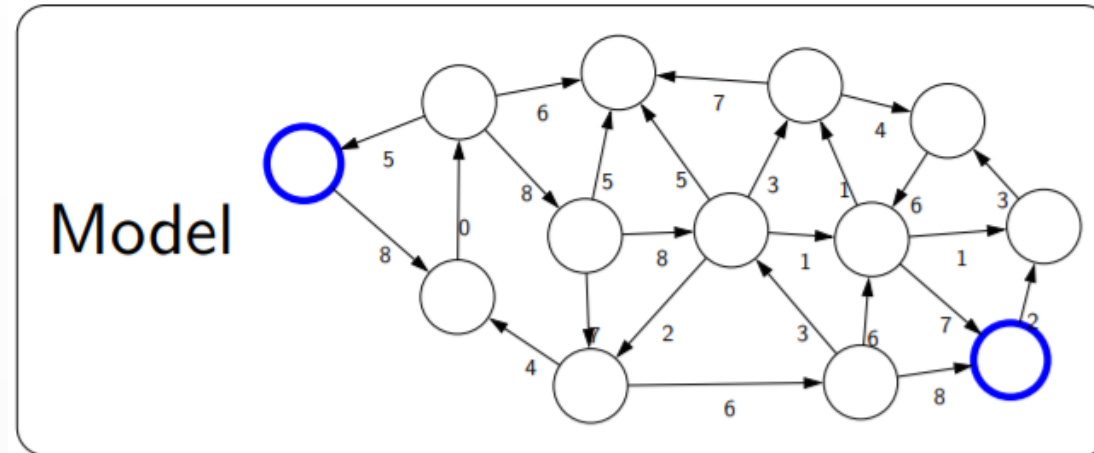
Modelling



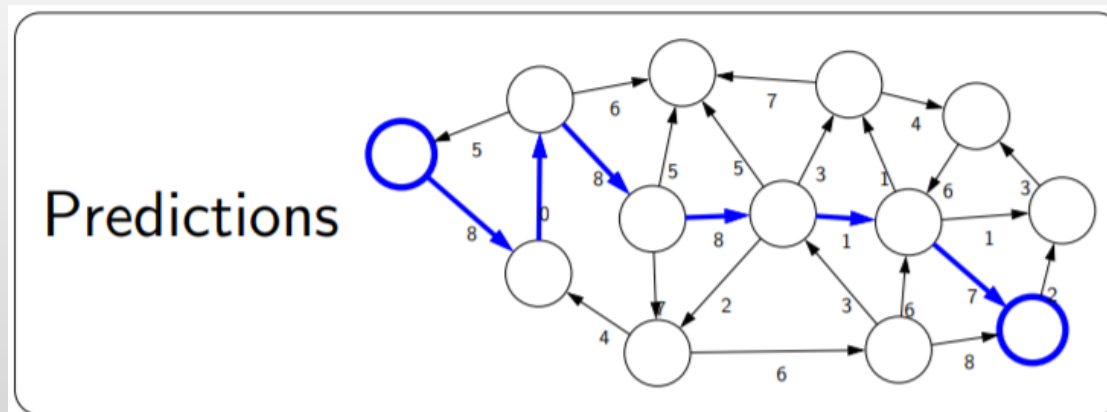
“All models are wrong, but
some models are useful.”

-- *George E.P. Box,*
statistician

AI Paradigm – Inference

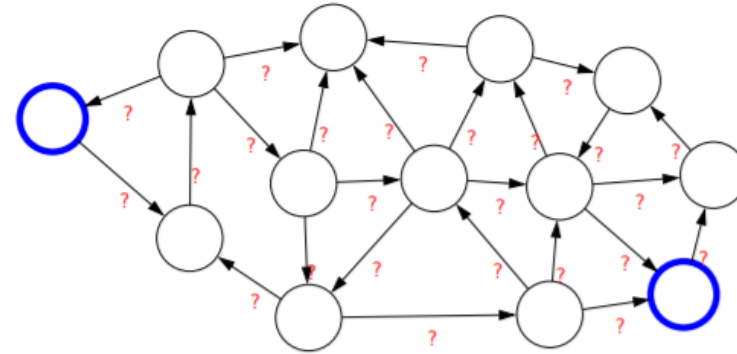


INFERENCE ↓



AI Paradigm – Learning

Model without parameters

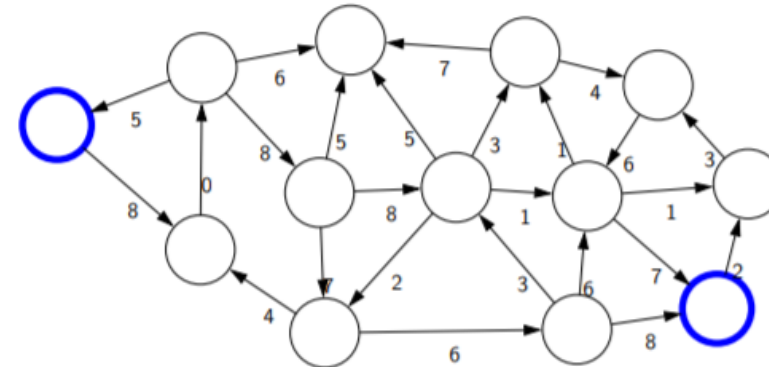


+DATA

LEARNING

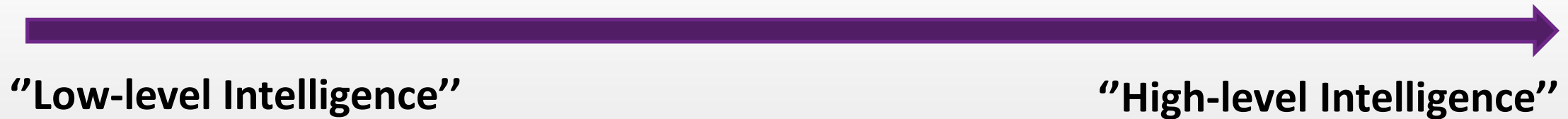


Model with parameters

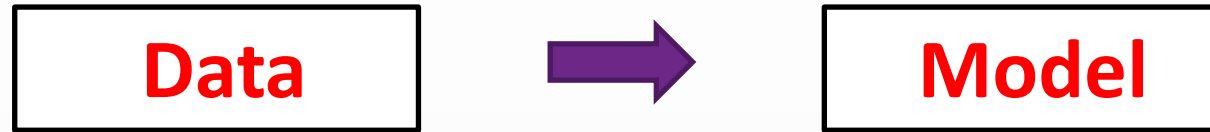


From low-level to high-level intelligence

Machine Learning

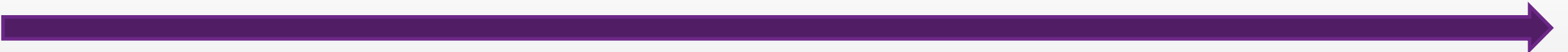


Machine Learning



- The main driver of recent successes in AI
- Move from "code" to "data" to manage the information complexity
- Requires a leap of faith: **generalization**

Reflex



“Low-level Intelligence”

“High-level Intelligence”

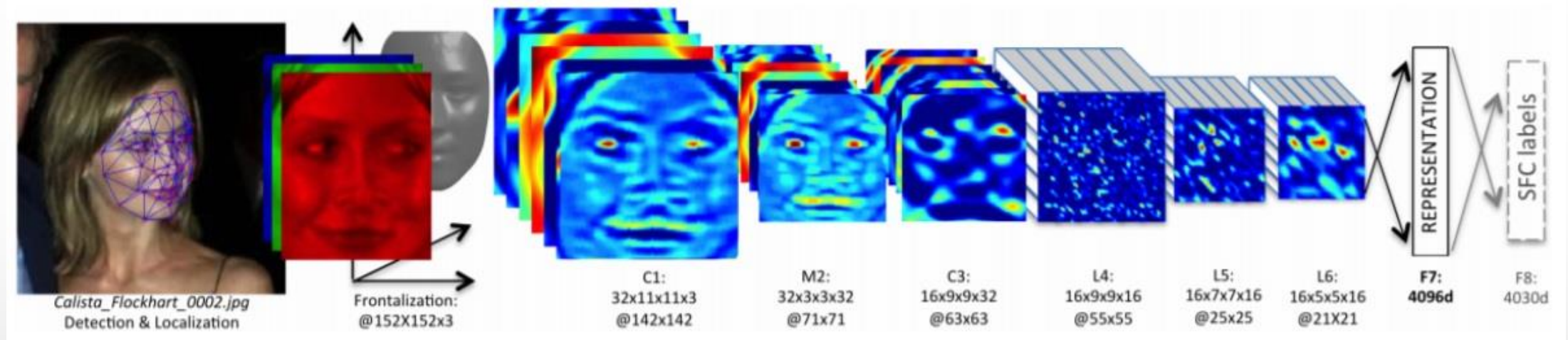
Machine Learning

What is this animal?



Reflex-based models

- Examples: linear classifiers, deep neural networks



- Most common models in machine learning
- Fully feed-forward (no backtracking)